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BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **SuryaSetu**

Team Leader Name : **Harsh Kawatra**

Problem Statement : **PS-15 · Nowcasting & Forecasting of Solar Flares from Aditya-L1 X-ray data**

Detect a flare the instant it begins — and warn minutes before its most damaging peak.

SoLEXS 2–22 keV · HELIOS 8–150 keV · LightGBM + MiniROCKET · TSS-driven · ₹0 on a laptop

Team Members

Team Leader

Name: Harsh Kawatra

College: Delhi Technological University (DTU)

Team Member-1

Name: Gursimran Kaur

College: Guru Gobind Singh Indraprastha University (GGSIPU)

Team Member-2

Name: —

College: —

Team Member-3

Name: —

College: —

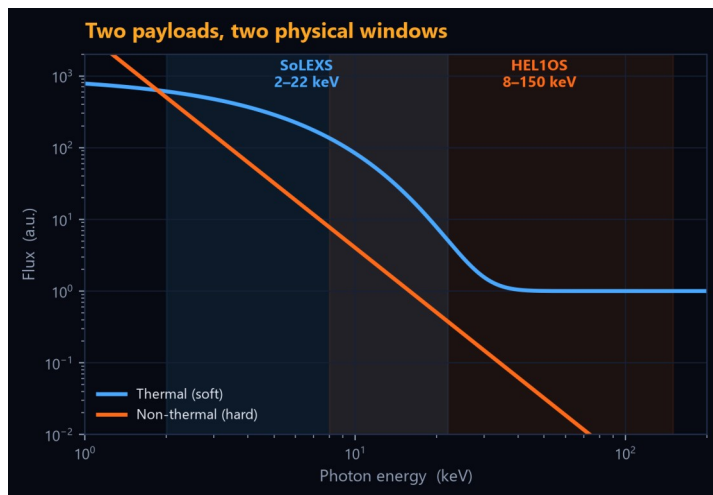
A two-person, two-institute team — full-stack + AI/ML engineering applied to ISRO's own solar mission.

Opportunity & USP — why SuryaSetu wins

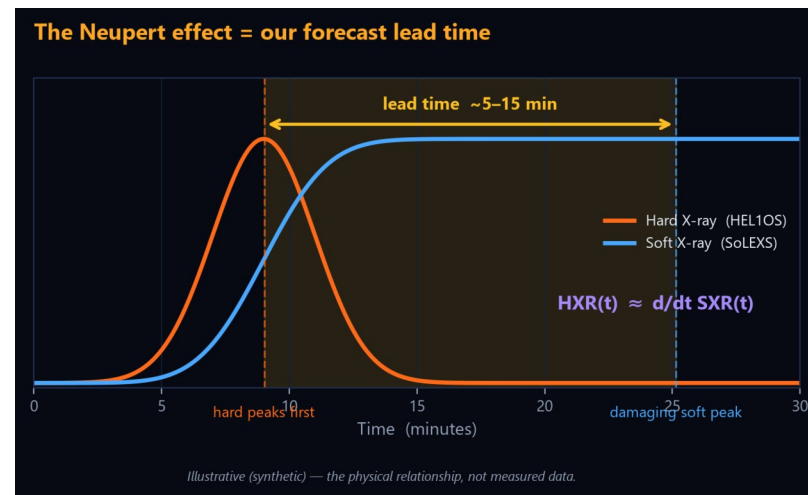
Most flare tools use a single channel from Earth orbit — and most “forecasts” are really just detections.

We are different on three axes:

- Dual-payload fusion — SoLEXS soft + HEL1OS hard, from the L1 vantage.
- Physics-grounded lead time — the Neupert effect, a real mechanism.
- Honest evaluation — TSS-driven, calibrated, time-blocked.



Two payloads = two physical windows (why fusion beats one channel)



Hard X-ray leads soft by 5–15 min = our forecast lead time

USP: the only dual-payload, L1-vantage system that turns the Neupert effect into calibrated, TSS-validated lead-time warnings — reproducible at ₹0 on a laptop.

Features offered by the solution

01

Automated nowcast catalogue

real-time soft+hard detection merged into one master catalogue · onset/peak/end · class · SNR

02

Calibrated forecast + lead time

$P(\text{flare} \leq N \text{ min})$ with calibrated confidence and a measured lead-time distribution

03

Live dashboard + visual alerts

dual curves · hard/soft ratio · gauge · green/amber/red banner that flips before the peak

04

Full A→X dynamic range

adaptive baseline + dual SoLEXS detectors catch faint A/B/C and giant M/X flares

05

Explainable alerts (SHAP)

every forecast states which precursor fired it — scientists can trust and audit it

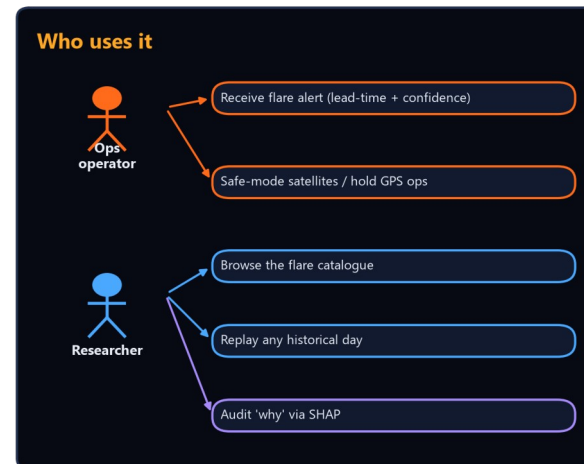
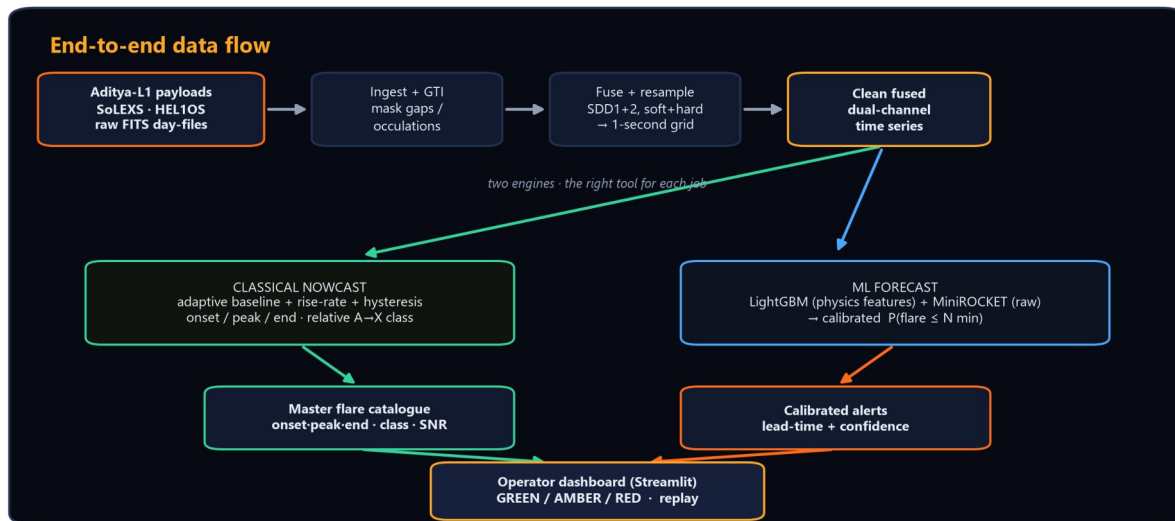
06

Offline & reproducible

no cloud, no cost, no proprietary software — end-to-end on one laptop from public data

Six capabilities — each mapped to an ISRO evaluation criterion: a master catalogue, calibrated lead-time forecasts, a live alerting dashboard, full A→X range, explainable alerts, and ₹0 reproducibility.

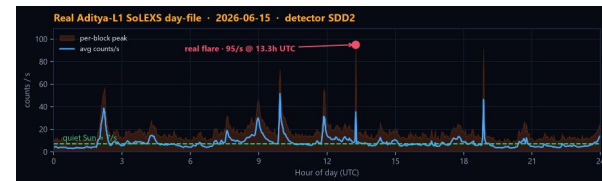
Process flow & use-case



Raw FITS → ingest + GTI → fuse to a 1-second grid → two parallel engines (classical nowcast || ML forecast) → catalogue + calibrated alerts → operator dashboard.

Two actors: an ops operator who acts on alerts, and a researcher who audits the catalogue.

Wireframe — the operator dashboard



Renders real Aditya-L1 data (2026-06-15)

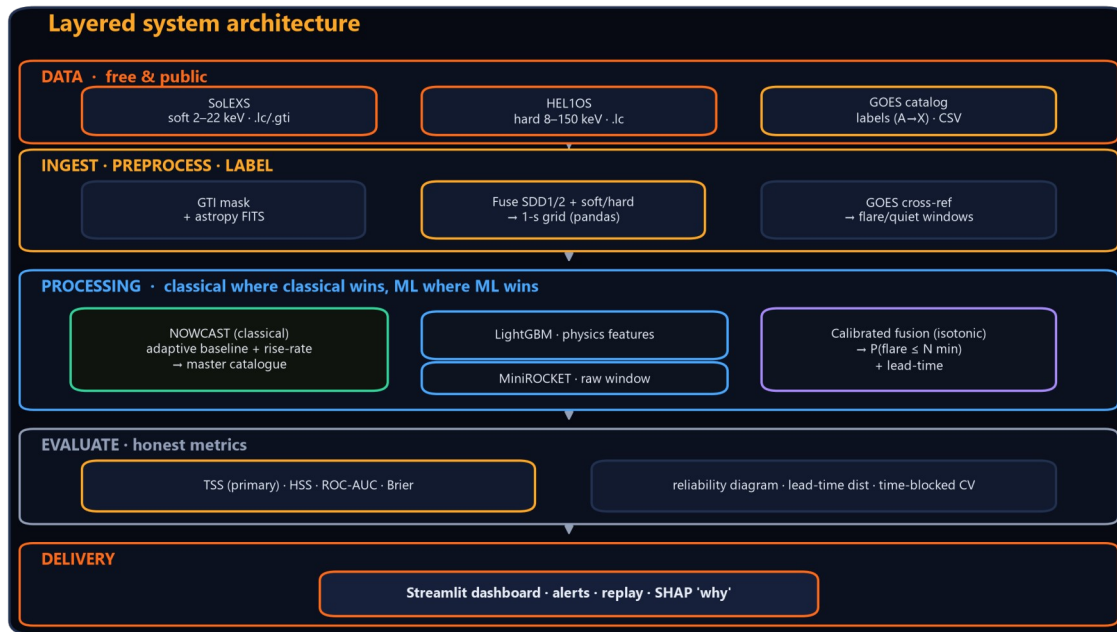
● **QUIET**
no flare activity

● **WATCH**
precursor (hard/soft ratio) rising

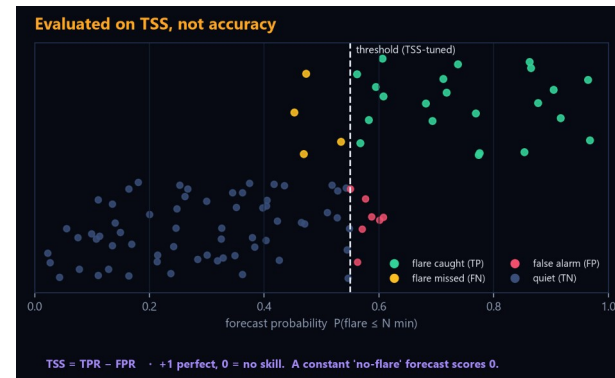
● **ALERT**
flare forecast — lead time + confidence shown

Alert banner spans the top; dual light curve + playhead centre; hard/soft-ratio precursor and a calibrated probability gauge right; the flare catalogue below. The banner flips before the soft peak.

Architecture of the proposed solution



Classical where classical wins, ML where ML wins. Evaluated on TSS (not accuracy); every alert is explainable.



Technologies used in the solution

Technology stack — every choice has a reason

LAYER	TOOL	WHY THIS TOOL	LICENSE
 FITS I/O	astropy	standard reader for SoLEXS/HEL10S products + GTI	BSD
 Numerics	numpy · pandas	time-series alignment, resampling, fusion	BSD
 Signal	scipy	detrend, smooth, derivative, adaptive baseline	BSD
 Forecaster A	LightGBM	fast, interpretable, sample-efficient on rare events	MIT
 Forecaster B	MiniROCKET (sktime)	CPU, deterministic, SOTA-ish on raw time series	BSD
 Calibrate/metric	scikit-learn	isotonic calibration · TSS/HSS/ROC/Brier · blocked CV	BSD
 Explainability	SHAP	per-alert precursor attribution — the 'why'	MIT
 Figures	matplotlib · plotly	light curves, reliability, lead-time plots	BSD
 Interface	Streamlit	fastest path to dual curves + alerts + replay	Apache
 Environment	Python 3.11 (uv)	prebuilt wheels; avoids the 3.14 astropy build break	PSF

Every layer, justified:

CPU-first — runs on an i5 / RTX 3050 / 16 GB laptop.

GPU optional, never required.

All open-source

(BSD / MIT / Apache / PSF).

Live build:

isro-hack-research.vercel.app

Estimated implementation cost

Prototype (our submission): total ₹0 — the only real input is engineering time.

Item	Cost	Note
Aditya-L1 SoLEXS + HEL1OS data	₹0	free · ISSDC PRADAN (registered)
GOES flare catalog (labels)	₹0	free · NOAA SWPC
Software stack	₹0	all open-source (BSD/MIT/Apache)
Compute	₹0	existing laptop — i5 / RTX 3050 / 16 GB
Storage	₹0	existing 256 GB drive (~5 GB used)
Hosting (demo)	₹0	Streamlit local / free HF Spaces
TOTAL (prototype)	₹0	only real input is engineering time

Smart data sampling

~70 GB
full Aditya-L1 archive



GOES-catalog targeting
flare + pre-flare + sampled quiet



~5 GB
downloaded · on a 256 GB drive

We never brute-force 70 GB —
labels point us to the 7% that matters.

Production / ISRO-scale deployment adds real cost — operational GPU/cloud compute (~₹1.5–2.5 L/yr), 24×7 ingest + redundant full-mission storage, monitoring/MLOps, and engineering staff. Scoped on selection.



SuryaSetu turns two ISRO payloads and one piece of solar physics into a calibrated, honestly-evaluated, zero-cost flare-warning system.

Live build · isro-hack-research.vercel.app

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THANK YOU